

Blockchain Agent

Mock Demo Ready

WorldLand Knowledge Wiki Agent

출처 검증 기반 논문 지식 그래프와
Blockchain Certificate Registry

Cheon Seungyeon · Final Project Presentation

Core idea

블록체인은 위키 저장소가 아니라
검증 통과 페이지의 인증서
레지스트리입니다 .

PASS → certificate
FAIL → local-only / SKIP_CHAIN

문제 정의 : LLM Wiki 는 유용하지만 위험합니다

Blockchain Agent · Mock Demo

LLM-generated knowledge bases need publication control.

Hallucination

근거 없는 claim 이 위키에
들어가면 노이즈 / 오정보가
됩니다 .

Passive hash log

단순 page hash 기록은 로컬
로그와 차별성이 약합니다 .

Need a gate

검증된 page 만 발행되어야
지식 품질과 보안이 함께
올라갑니다 .

Design question: “WorldLand 에 기록이 있다는 사실이 무엇을 증명하는가 ?”

답 : 이 page hash 는 EvalAgent 의 source-grounded verification 을 통과한 certificate 이다 .

교수님 피드백 반영 : Hash Logger → Certificate Registry

Blockchain Agent · Mock Demo

The blockchain role changed from recording to gating.

Before: Passive Hash Logger

- 위키 page 생성 후 page_hash 기록
- hash + timestamp + agent name
- 문제 : 로컬 로그와 본질적 차이가 약함

Local log 와 비슷한 수준

After: Publication Certificate Registry

- EvalAgent 가 claim groundedness 검증
- $\text{score_bps} \geq \text{threshold_bps}$ 일 때만 certify
- FAIL 은 SKIP_CHAIN, tx_hash=None

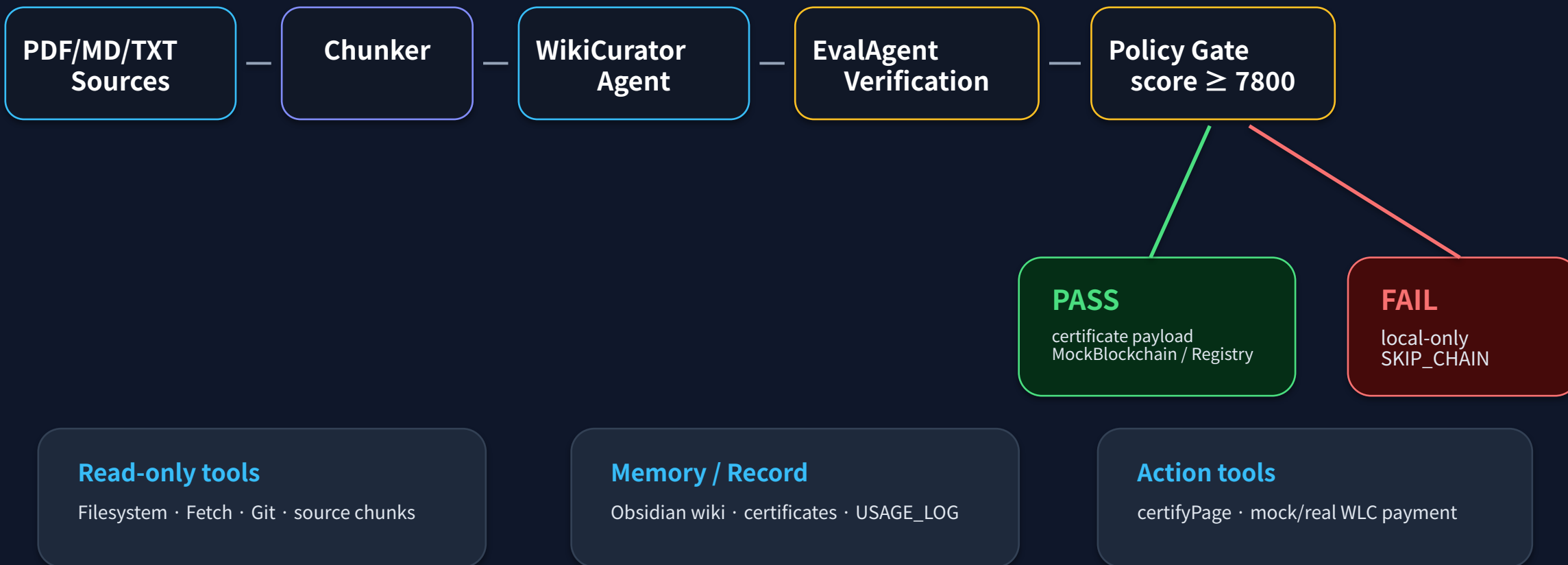
온체인 기록 = 검증 통과 인증서

Professor feedback implemented

전체 Agentic Workflow

Blockchain Agent · Mock Demo

Goal → Input → Tools → Check → Record → Human Review → Output



Publication Gate: 온체인 기록의 의미

Blockchain Agent · Mock Demo

A certificate exists only if the page passes source-grounded verification.

```
if score_bps >= threshold_bps and not
critical_unsupported:
    tx_hash = certifyPage(page_hash,
cert_hash, score_bps)
else:
    tx_hash = None
    log('SKIP_CHAIN')
```

PASS

- certificate JSON 생성
- page_hash + cert_hash
- Mock/real tx 생성

FAIL

- local evaluation log
- SKIP_CHAIN
- tx_hash=None

Key phrase: “The absence of a certificate is the rejection.”

실패한 page 는 reject event 조차 온체인에 남기지 않습니다 .



WorldLand explorer / real tx screenshot
(optional after WLC integration)

EvalAgent Groundedness Scoring

Claim-level verification with basis points.

Scoring rule

SUPPORTED	1.0	근거 충분
PARTIAL	0.5	부분 근거
UNSUPPORTED	0.0	근거 없음
score_bps = average × 10000		

Sample evaluation fixture at threshold_bps = 7800

sample	score	critical	decision
strong	9200	False	PASS
good	8700	False	PASS
borderline	8100	False	PASS
partial	7600	False	FAIL
weak	7200	False	FAIL
critical hallucination	0	True	FAIL

Gate behavior 는 fixture 로 , large batch 는 scalability evidence 로 설명합니다 .

Cross-linking: 후보 생성과 최종 채택을 분리

Blockchain Agent · Mock Demo

Cosine similarity is candidate generation only.

TF-IDF / Cosine

candidate links only

EvalAgent

accept / reject with evidence

Obsidian Page

accepted links only

ACCEPT example

State Transition → Gas
Reason: source evidence shows transaction execution consumes gas.

REJECT example

State Transition → Polymarket
Reason: semantic similarity but no source-backed relation.



Obsidian graph / generated wiki page screenshot

Latest Demo Run Summary

Blockchain Agent · Mock Demo

Reproducible mock mode for presentation.

```
python demo/demo_pipeline.py --threshold-bps 7800 --llm-provider mock --mock-chain --mock-payment --append-usage-log
```

109
source files

1328
chunks

110
pages

109 / 1
PASS / FAIL

109
mock certs

545
mock payments

Important framing Large batch 는 scalability/practicality evidence 입니다 . Publication gate behavior 는 sample fixture 와 FAIL evidence 로 설명합니다 .

PASS vs FAIL Evidence in Logs

Blockchain Agent · Mock Demo

The key professor-feedback evidence.

PASS → CERTIFIED

```
CERTIFIED | 0xmock_cert_da5400... | 3D Diffuser  
Actor | score_bps=10000 threshold_bps=7800
```

- evaluation certificate exists
- mock certificate tx exists
- WorldLand interface can be swapped to real mode later

FAIL → LOCAL ONLY

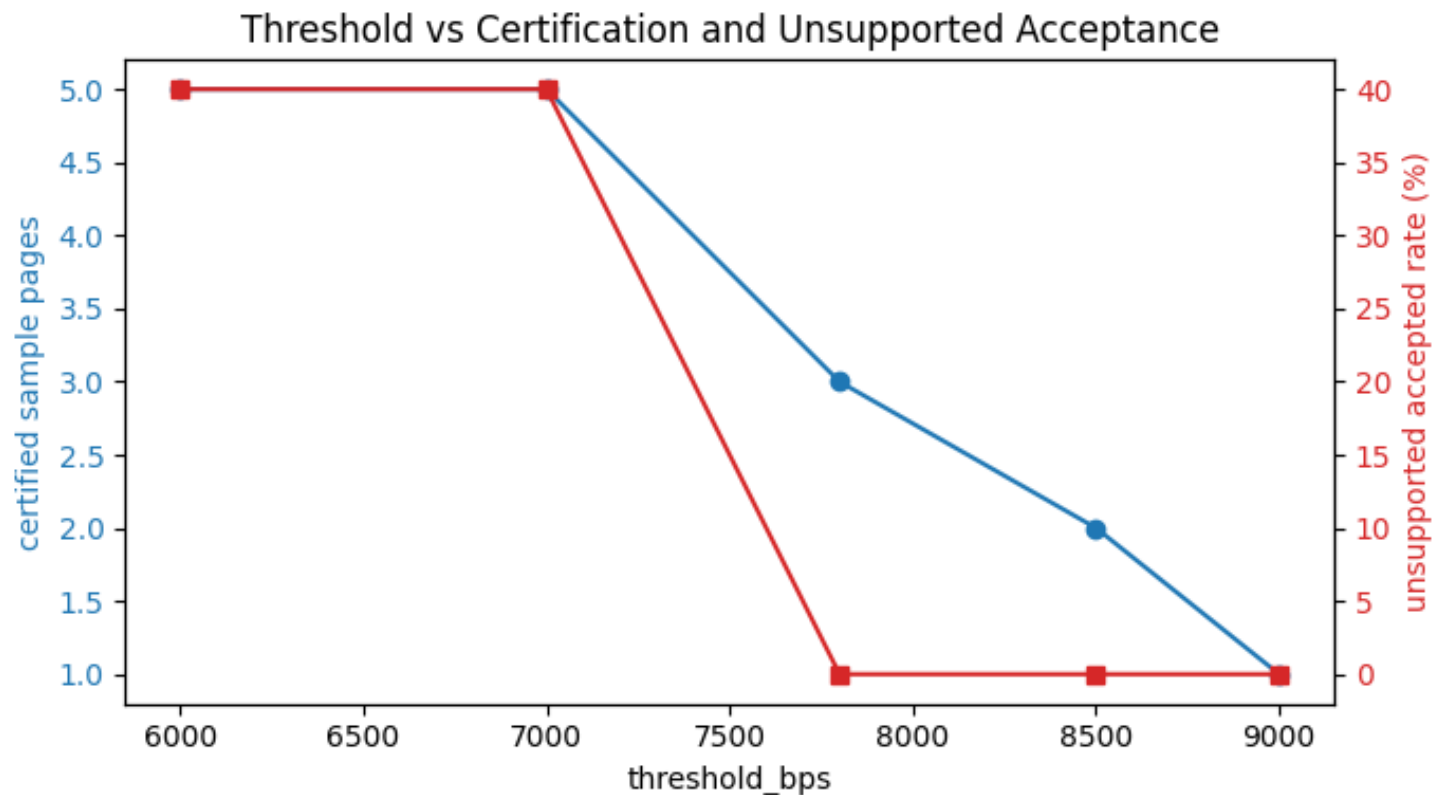
```
SKIP_CHAIN | tx_hash=None | Hallucinated Failure  
Fixture | score_bps=0 threshold_bps=7800  
critical=True  
FAIL_LOCAL_ONLY | ... | chain_action=SKIP_CHAIN
```

- no blockchain call
- no mock certificate event
- absence of certificate = rejection

Threshold Trade-off: Quality Gate Sensitivity

Blockchain Agent · Mock Demo

Fixture-based sensitivity test.



Interpretation

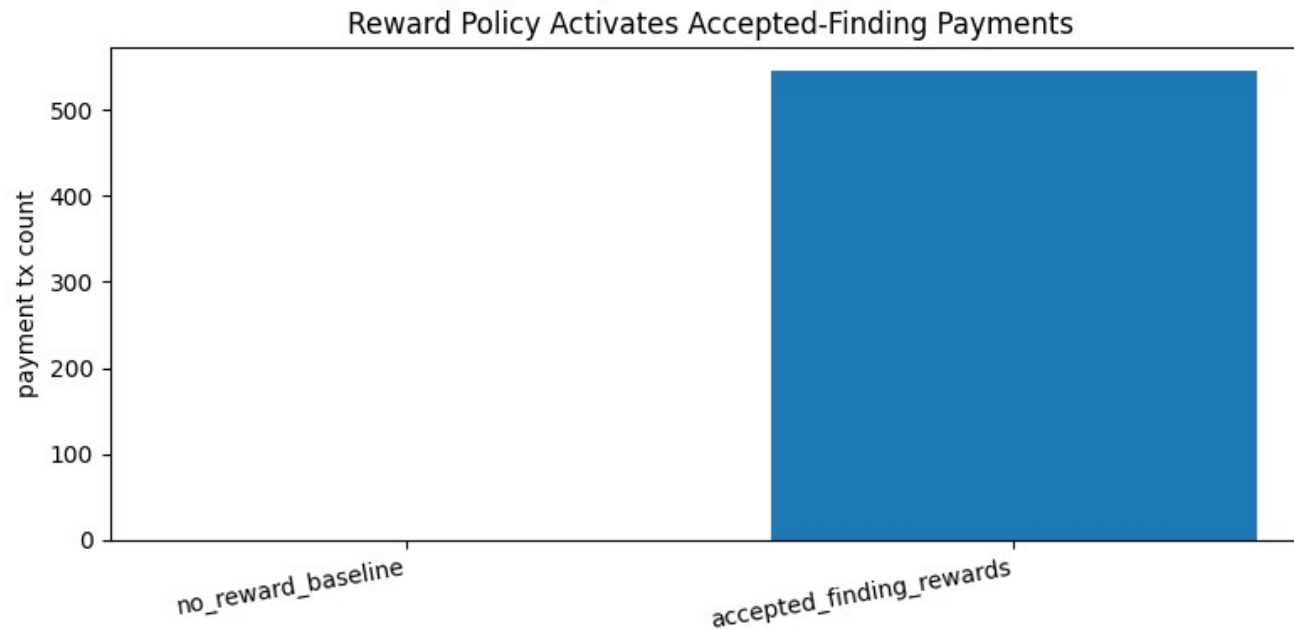
- threshold 가 올라가면 certified sample pages 가 감소
- unsupported accepted rate 가 40% → 0% 로 감소
- large batch 는 mock evaluator 특성상 대부분 10000 점
- 따라서 gate 민감도는 fixture 로 설명

At threshold 7800: sample 3 PASS / 3 FAIL

PayAgent: Accepted Finding Reward Flow

Blockchain Agent · Mock Demo

Reward is paid per accepted EvalAgent finding, not per passed page.



Reward policy

- no_reward_baseline: 0 payments
- accepted_finding_rewards: 545 mock payments
- total mock WLC: 5.45
- 이 그래프는 quality improvement 증거가 아니라 payment flow 증거

```
0xmock_pay... | finding=f... |  
amount_wlc=0.01
```

실제 WLC payment 는 아직 없습니다 .

WorldLand / WLC Integration Plan

Blockchain Agent · Mock Demo

Mock mode today; real tx can be added after WLC and wallet setup.

What goes on-chain?

- page_hash
- cert_hash
- score_bps / threshold_bps
- page_id / evaluator / timestamp
- No full paper text or private notes

WLC use cases

- contract deploy gas
- certifyPage tx gas
- accepted finding reward
- retry / demo buffer

Recommended request: 50 WLC

Current status

- real WorldLand tx: none
- real WLC payment: none
- MOCK_CHAIN /
MOCK_PAYMENT
- mock prefixes: 0xmock_cert_,
0xmock_pay_

Do not present mock tx as real tx

Privacy & Security Design

Blockchain Agent · Mock Demo

Local content, minimal on-chain metadata, explicit approval for risky actions.

Privacy by default

- PDFs and wiki content stay local
- Only hashes/cert metadata on-chain
- No private notes on WorldLand

Secret handling

- No .env committed
- Private keys only from env vars
- Never print wallet key

Threat controls

- FAIL → SKIP_CHAIN only
- Mock tx prefixes separate reality
- Real payment blocked without confirmation

Security takeaway: certificate registry proves a page version passed verification without exposing source content.

사진 / 영상 삽입 위치

Blockchain Agent · Mock Demo

Blank placeholders for screenshots or demo video.



Terminal demo output
python
demo/demo_pipeline.py ...



Obsidian vault graph
or generated wiki page



Evaluation certificate JSON
claim checks + score_bps



CERTIFICATE_LOG.md
CERTIFIED + SKIP_CHAIN
lines



PAYMENT_LOG.md
accepted finding mock
reward



Demo video thumbnail
≤ 5 min screen capture

Tip

실제 WorldLand explorer screenshot 은 WLC 수령 후 real tx 가 생겼을 때만 추가합니다. 현재 발표에서는 “real tx 없음” 을 명확히 표시합니다.

Limitations & Final Takeaway

Blockchain Agent · Mock Demo

Honest boundaries and next steps.

Current limitations

- Mock evaluator: not full LLM claim verification
- No real WorldLand tx yet
- No real WLC payment yet
- Reward graph shows payment flow, not proven quality improvement
- Large batch scores are optimistic under mock mode

Next steps

- Add Ollama/OpenAI EvalAgent mode
- Deploy PageCertificateRegistry on WorldLand
- Use WLC for deploy + certifyPage + accepted finding rewards
- Add block explorer proof and read-only tx receipt verification

Final takeaway: WorldLand is not a passive hash log. It is a publication certificate registry.